

Intellectual Property Acknowledgement

You, the author of this white paper, agree that you have not used any copyrighted material in this white paper or, have appropriately indicated the copyright of the respective object. You further agree that you have not used any other intellectual property of any third party in this white paper or, have appropriately indicated the intellectual property of the respective object.

You confirm, that consistent with the terms of your employment with Accenture, that any rights, title and interest whatsoever, including, but not limited to, patents, copyright, trade secret and design rights, mask rights, whether register able or not, arising or created as a result of the development of and/or the application of any tangible or intangible work product or materials produced by you during or as a consequence of your employment (including in this white paper), whether alone or in conjunction with others and whether during normal working hours or not, including, but not limited to, any invention, design, discovery, improvement, computer program, documentation, or other material (including in this white paper) which you conceive, discover or create during or in consequence of employment hereunder ("Work Product") shall belong exclusively to Accenture and its affiliates. You hereby convey ownership in such rights, title and interest to Accenture and its affiliates upon inception or development.

All Work Product shall constitute a work(s) made for hire under all copyright acts. To the extent that any Work Product does not constitute a work made for hire under the foregoing laws, you hereby irrevocably assign all worldwide right, title, and interest (including without limitation, patents, copyright, trade secret, trademarks, design rights, contract and licensing rights) in such Work Product to Accenture and its affiliates. You retain no rights to use the Work Product and agree not to challenge the validity of Accenture's and its affiliates' ownership in the Work Product. You hereby forever waive all moral rights in the Work Product and any results or proceeds there from, even if after expiration or termination of your employment hereunder. If you agree that you will not violate or attempt to violate the intellectual property rights (IPR), interests or title of any third party. Accenture shall be entitled to immediate injunctive or similar relief upon a potential or actual breach of this Section by you. You agree to cooperate with Accenture and its affiliates as is necessary and provide any necessary documentation consistent with and in furtherance of the rights conveyed hereunder.

By submitting this white paper, you agree that you have read and understood the disclaimer set out above and that you understand that it may affect your rights and agree to be bound by these terms.

Submitter Credentials

Name of the Employee	<u>VIVEK RANJAN</u>
Employee SAP ID	<u>10809507</u>
Enterprise ID	<u>v.a.ranjan</u>
Current Project Name	<u>VATS</u>
Topic Category (Please select the category from the list in the announcement mailer)	<u>Cloud Testing</u>

A large, stylized graphic with the words "Cloud" in yellow and "Testing" in light blue, set against a background of soft, white clouds.

Going beyond Conventional Software Testing: Cloud Testing

ABSTRACT

Software testing is an important part of software engineering life cycle. Testing requires add-on resources that are often not readily available, contributing to an inefficient testing process. Running large no. of test cases can consume a lot of time and resources, often precluding their use in an interactive setting. Cloud computing has changed the way of obtaining computing resources, and also has given a new direction to manage and deliver computing services, technologies, and solutions. Cloud computing creates an opportunity that offers testing as a service (TaaS) for Software as a service (SaaS) and clouds. This lead to a new phase shift in conventional testing thereby identifying new issues, challenges and needs in software testing, particular in testing clouds and cloud-based applications. This paper gives a comprehensive view on cloud testing and cloud-based application testing that can be the possible answers to the common questions in conventional testing & clouds. Also, it examines the major issues, challenges, and needs in testing cloud-based software applications.

Keywords - Cloud testing, cloud-based software testing, testing cloud services, TaaS, SaaS, IaaS, and PaaS.

1. INTRODUCTION

Cloud computing received significant role in the way computation and services to customers, For example, it changes the way of providing and managing computing resources, such as CPUs, databases, and storage systems. Today, leading players, such as Amazon, Google, IBM, Microsoft, and Salesforce.com offer their cloud infrastructure for services. Cloud computing is a business and economical model. This model has been successfully deployed and executed for various commodities since its inception, but in recent years it has been more into IT products & services. Consider the example of automobile to understand the usage of Cloud Computing. Take the case of car versus taxi cab. Both are vehicles for carrying people from one place to another. The difference is the business model for the services provided by them. Owner of the car pay for the fuel & maintenance, and the car offer services solely to the owner. On the other hand, services provided by a taxi cab are “Travel as a Service”. The taxi driver owns the cab. As a customer, one has to pay to travel to use, there is no need of maintenance. The responsibility lies with the driver. This is why the Cloud is synonymous with “On Demand”. You pay only on demand (when to require) it.



Fig 1. Taxi as “Travel as a Service”

IT industry over a period of time offers wide range of services – on demand. We often come across “Games as a service”, “Java as a service”, “Storage as a service” and even more. Cloud computing received significant attention

recently as it changes the way computation and services to customers, For example, it changes the way of providing and managing computing resources, such as CPUs, databases, and storage systems. Today, leading players, such as Amazon, Google, IBM, Microsoft, and Salesforce.com offer their cloud infrastructure for services. A recent study of Market Research Media forecasts that U.S. government spending on cloud computing is entering an explosive growth phase at about 40% CAGR over the next six years. Expenditure will pass \$7 billion by 2015. Merrill Lynch estimates that within the next five years, the annual global market for cloud computing will surge to \$95 billion. Cloud computing not only brings new business opportunities, but also causes some major impacts on software testing and maintenance. A major impact is known as testing as a Service (TaaS) in Clouds. TaaS cloud infrastructures is considered as a new business and service model, in which a provider undertakes software testing activities of a given application system in a cloud infrastructure for customers as a service based on their demands. Although there are many published papers discussing cloud architectures, technologies, and models, design, and management, cloud testing and TaaS are still new subjects in software testing community. Hence, test engineers and quality assurance managers encountered many issues and challenges in testing modern clouds and cloud-based applications.

Typical questions are listed below.

- What is cloud testing? And what are its special test process and scope, requirements and features?
- What types of cloud testing, environments, and forms do we need to perform for SaaS/clouds and cloud-based applications?
- What are the major differences between conventional software testing and cloud-based software testing?
- What are the special requirements and distinct features of cloud-based software testing?
- What are the special issues, and challenges, and needs in cloud testing?
- What are the current practice, tools, and major players?

This paper is written to attempt to answer these questions. The paper introduces basic concepts about cloud testing, including its scope, objectives, distinct requirements, features, and benefits. Also a comparative view towards conventional testing and cloud testing is also considered

2. UNDERSTANDING CLOUD TESTING

Cloud Computing provides a cost-effective and flexible means through which scalable computing power and diverse services (computer hardware and software resources, networks and computing infrastructures), diverse application services, business processes to personal intelligence and collaboration are delivered as services to large-scale global users whenever and wherever they need. Cloud computing is the next stage of the Internet evolution. A typical cloud must have several distinct properties: elasticity and scalability, multi-tenancy, self-managed function capabilities, service billing and metering functions, connectivity interfaces and technologies. In addition, a cloud supports large scale user accesses at distributed locations over the Internet, offers on-demand application services at any time, and provides both virtual and/or physical appliances for customers.

There are three types of clouds:

- a) Private clouds, which are internal clouds based on a private network behind a firewall;
- b) Public clouds, which are the clouds with public accessible services over the Internet; and
- c) Hybrid clouds, which are made of different types of clouds, including public and private clouds.

2.1 WHAT IS CLOUD TESTING?

Cloud Testing is a form of software testing in which Web applications that leverage Cloud computing environments ("cloud") seek to simulate real-world user traffic as a means of load testing and stress testing web sites.

The ability and costs to simulate Web traffic for software testing purposes has been an inhibitor to overall Web reliability." In short, cloud-based software testing refers to testing and measurement activities on a cloud-based environment and infrastructure by leveraging cloud technologies and solutions. It has three major objectives.

- To assure the quality of cloud-based applications deployed in a cloud, including their functional services, business processes, and system performance as well as scalability based on a set of application-based system requirements in a cloud.
- To validate software as a service (SaaS) in a cloud environment, including software performance, scalability, security and measurement based on certain economic scales and pre-defined SLAs.

- To check the provided automatic cloud-based functional services, for example auto-provisioned functions.
- To test cloud compatibility and inter-operation capability between SaaS and applications in a cloud infrastructure, for example, checking the APIs of SaaS and their cloud connectivity to others.

2.2 WHY IS CLOUD TESTING IMPORTANT?

Comparing with current software testing, cloud-based testing has several unique advantages listed below.

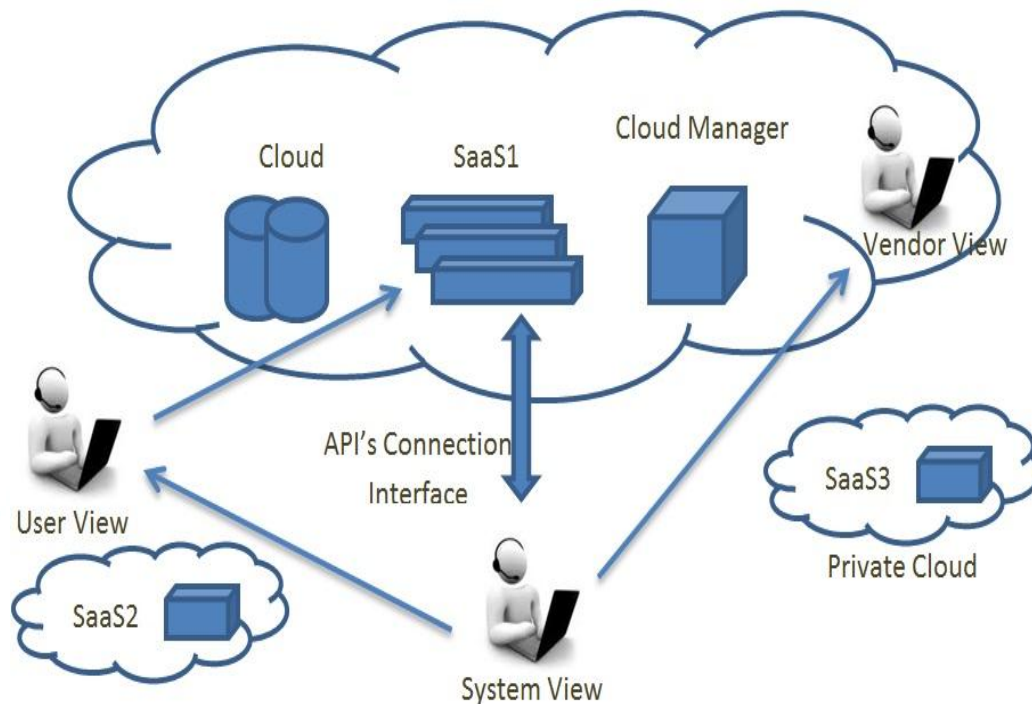


Figure 2: Different Views for Cloud-based Software Testing

- ✓ Reduce costs by leveraging with computing resources in clouds – This refers to effectively using virtualized resources and shared cloud infrastructure to eliminate required computer resources and licensed software costs in a test laboratory.
- ✓ Take the advantage of on-demand test services (by a third-party) to conduct large-scale and effective real-time online validation for internet-based software in clouds.
- ✓ Easily leverage scalable cloud system infrastructure to test and evaluate system (SaaS/Cloud/Application) performance and scalability.
- ✓ Reduce its capital and licensing expenses as much as 50% to 75% using virtualized resources.
- ✓ Reduce operating and labor costs as much as 30% to 50% by automating development and testing resource provisioning and configuration.
- ✓ Shorten its development and testing setup time from weeks to minutes.
- ✓ Improve product quality and reduce the detected defects by as much as 15% to 30%.

2.3 FORMS OF CLOUD-BASED SOFTWARE TESTING

There are four different forms of cloud-based software testing. Each of them has different focuses and objectives.

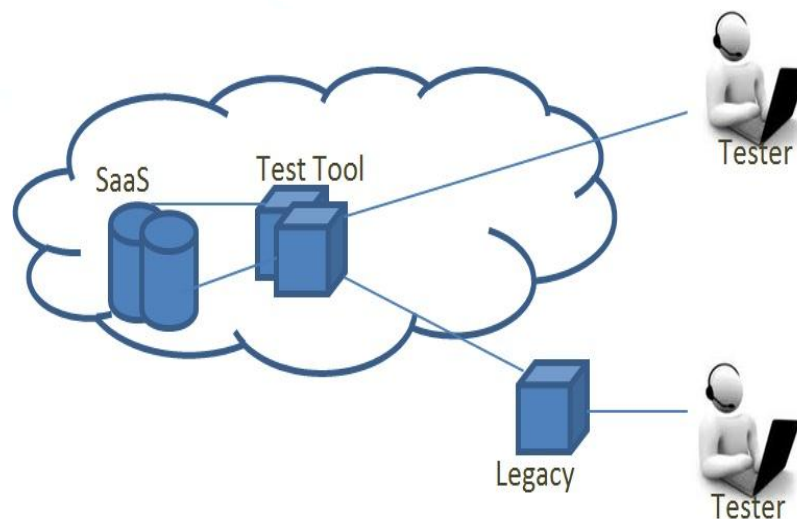
- Testing a SaaS in a cloud – It assures the quality of a SaaS in a cloud based on its functional and non-functional service requirements.

- Testing of a cloud – It validates the quality of a cloud from an external view based on the provided cloud specified capabilities and service features. Cloud and SaaS vendors as well as end users are interested in carrying on this type of testing.
- Testing inside a cloud - It checks the quality of a cloud from an internal view based on the internal infrastructures of a cloud and specified cloud capabilities. Only cloud vendors can perform this type of testing since they have accesses to internal infrastructures and connections between its internal SaaS(s) and automatic capabilities, security, management and monitor.
- Testing over clouds – It tests cloud-based service applications over clouds, including private, public, and hybrid clouds based on system-level application service requirements and specifications.

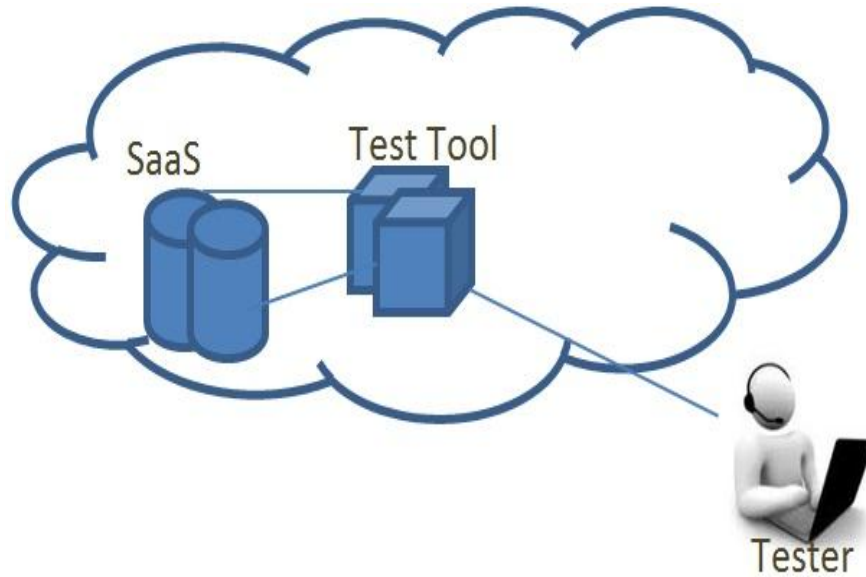
Figure 2 shows three different views of software testing in a cloud environment. The first is the vendor view, which presents the testing view from the engineers of a cloud vendor. They perform vendor-oriented software testing tasks. The next is the user view, which presents the consumer-oriented testing view from cloud-based application users through web-based user interfaces. They conduct testing and QA jobs to assure the quality of provided application services in a system-oriented test view in a given cloud infrastructure where different cloud-based applications may interact with each other. They need to perform different testing tasks to assure the quality of the cloud-based application systems over clouds, such as cloud-based application integration, end-to-end system function testing, system performance and scalability over different clouds.

Figure 3, shows three types of cloud environment:

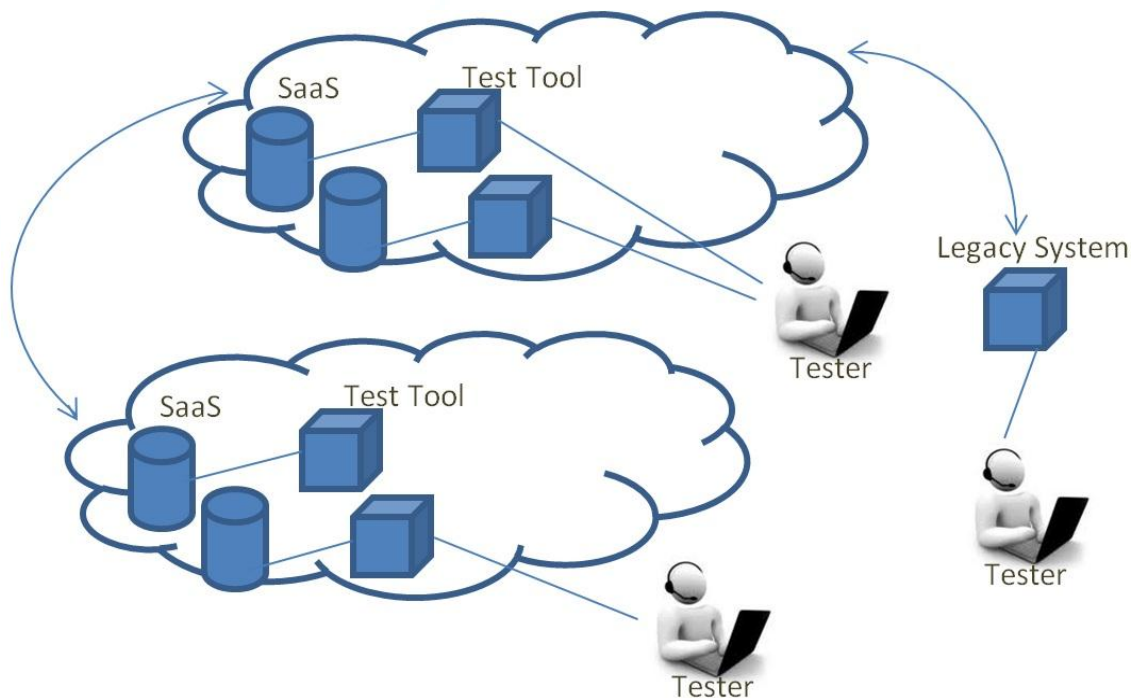
- A cloud-based enterprise test environment, in which application vendors deploy web-based applications in a cloud to validate their quality in a cloud infrastructure.
- A private/public cloud test environment, in which vendors deploy SaaS applications SaaS in a private (or public) cloud to validate their quality.
- A hybrid cloud test environment, in which vendors deploy cloud-based applications on a hybrid cloud infrastructure to check their quality.



(a) Private / Public Cloud Test Environment



(b) Cloud based Enterprise Test Environment



(c) Hybrid Cloud Test Environment

Figure 3: Different Cloud Test Environment

Different tasks performed in three types of cloud testing below.

✓ **Cloud/SaaS-oriented testing** - This type of testing activities usually is performed inside a cloud by engineers of cloud/SaaS vendors. The primary objective is to assure the quality of the provided service functions offered in a cloud (or a SaaS program). These engineers must go through unit testing, integration, system function validation and regression testing, as well as performance and scalability evaluation. Since clouds and SaaS usually provide certain service APIs and connectivity interfaces to their customers, it is required task for engineers to validate these APIs and connectivity in a cloud environment. In addition, testing cloud-based or SaaS-based security services and functional features must be tested. Furthermore, performance testing and scalability evaluation in a cloud is very important and critical to cloud/SaaS vendors because this assures the quality of cloud elasticity to support SaaS and cloud services inside a cloud.

✓ **Online-based application testing on a cloud** - This type of testing activities usually is performed to check online application systems on a cloud by using with cloud-based large-scale traffic and user accesses. This is a common usage of cloud technology to help current online application vendors to conduct online-based system function testing and performance evaluation on a cloud by taking the advantage of cloud environment so that diverse and scalable computing resources in a cloud can be used without using any in-house test laboratory. When applications are connected with legacy systems, the quality of the connectivity between the legacy systems and the under-test application deployed on a cloud must be validated.

✓ **Cloud-based application testing over clouds** - This type of testing refers to the engineering activities performed to assure the quality of a cloud-based application crossing different clouds. When applications are developed to be deployed and executed over different clouds, new testing tasks are needed to assure its quality. Unlike the previous two types, the primary testing objective here is to assure the quality of the end-to-end application over clouds.

This suggests that the system-level integration, function validation, performance evaluation, and scalability measurement must cope with different cloud technologies. This definitely complicates the tasks for checking system compatibility, interoperability, and connectivity between different clouds.

3. CLOUD TESTING

Unlike testing conventional web-based software, testing clouds and cloud-based software has several unique testing quality assurance objectives, requirements, and distinct features.

3.1 Types of Cloud Testing

Various types of testing required for a cloud setup, some of the types of cloud testing mention in below:

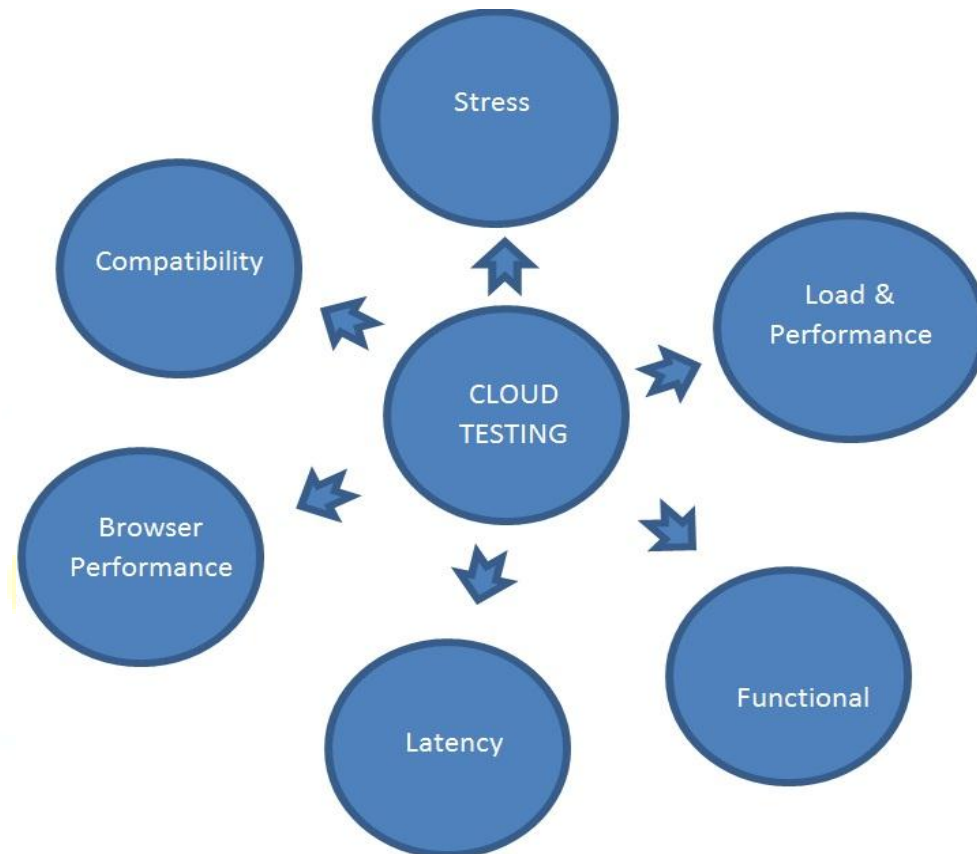


Figure 4: Types of Cloud Testing

3.1.1 Stress test

Stress Test is used to determine ability of application to maintain a certain level of effectiveness beyond breaking point. It is essential for any application to work even under excessive stress and maintain stability. Stress testing assures this by creating peak loads using simulators. But the cost of creating such scenarios is enormous. Instead of investing capital in building on-premise testing environments, cloud testing offers affordable and scalable alternative.

3.1.2 Load & Performance Test

Load testing of an application involves creation of heavy user traffic, and measuring its response. There is also a need to tune the performance of any application to meet certain standards. Finding out thresholds, bottlenecks & limitations is a part of performance testing. For this, testing performance under a particular workload is necessary. By using cloud testing, it is easy to create such environment and vary the nature of traffic on-demand. This effectively reduces cost and time by simulating thousands of geographically targeted users.

3.1.3 Functional Testing

Functional testing of both internet and non-internet applications can be performed using cloud testing. The process of verification against specifications or system requirements is carried out in the cloud instead of on-site software testing.

3.1.4 Compatibility Testing

Using cloud environment, instances of different Operating Systems can be created on demand, making compatibility testing effortless.

3.1.5 Browser Performance Testing

To verify application's support for various browser types and performance in each type can be accomplished with ease. Various tools enable automated website testing from the cloud.

3.1.6 Latency Testing

Cloud testing is utilized to measure the latency between the action and the corresponding response for any application after deploying it on cloud.

3.2 Features in Cloud Testing

There are four new requirements and features in cloud testing.

❖ Cloud-based testing environment

This refers to use a selected cloud infrastructure (or platform) as a base to form a test bed equipped with diverse and scalable computing resources, system infrastructures, and licensed tools, which are allocated using auto-provision based on static/dynamic requests. Both virtual and physical computing resources can be included and deployed inside.

❖ Service-level-agreements (SLAs)

In cloud computing, all clouds, SaaS, and applications usually provide diverse services to their end users and customers with well-defined service-level-agreement. Naturally, these agreements will become a part of testing and quality assurance requirements, such as system reliability, availability, security, and performance agreements.

❖ Price models and service billing

Since utility computing is one of basic concepts and features in cloud computing, so price models and utility billing becomes basic parts and service for testing as a service. In other words, required computing resources and infrastructures (including tools), and testing task services will be charged based on pre-defined cost models and • Large-scale cloud-based data and traffic simulation - Applying and simulating large-scale online user accesses and traffic data (or messages) in connectivity interfaces is necessary in cloud testing, particularly in system-level function validation and performance testing.

3.3 Testing as a Service (TaaS)

There are several distinct features in cloud testing. One of them is testing as a service (TaaS). This is an innovative concept, and it refers to providing static/dynamic on-demand testing services in/on/over clouds for the third-parties at any time and all time (365/7/24).

One of the primary objectives is to reduce the IT budget of businesses to focus their core businesses by outsource software testing tasks to a third party using TaaS service model. TaaS involves the on-demand test execution of well-defined suites of test material, generally on an outsourced basis. The execution can be performed either on client site or remotely from the outsourced providers test lab/facilities.

TaaS has received wide attention due to its advantage in its scalable testing environment, cost reduction, utility-based service models, and on-demand testing services.

The work-flow of TaaS includes the following major TaaS service capabilities.

- TaaS process management, which offers test project management and process control.
- QoS requirements management, which supports book keeping and modeling of software testing and QoS requirements, including quality assurance modeling.
- Test environment service, which provides on-demand test environment services to establish the required virtual (or physical) cloud-based computing resources and infrastructures, as well as the necessary tools.
- Test solution service, which offers diverse systematic testing solutions (such as, test modeling and test methods), and test-ware generation and management services.
- Test simulation service, which establishes on-demand test simulation environments with selected facilities (such as tools), and supports the necessary test data/message generation.
- On-demand test service, which provides on-demand test execution services based on selected schedules and test wares.
- Tracking and monitor service, which allows test engineers to track and monitor diverse program behaviors at different levels in/on/over clouds for the testing purpose.
- TaaS pricing and billing, which enables TaaS vendors to offer customers with selectable testing service contracts based pre-defined pricing models, and billing service.

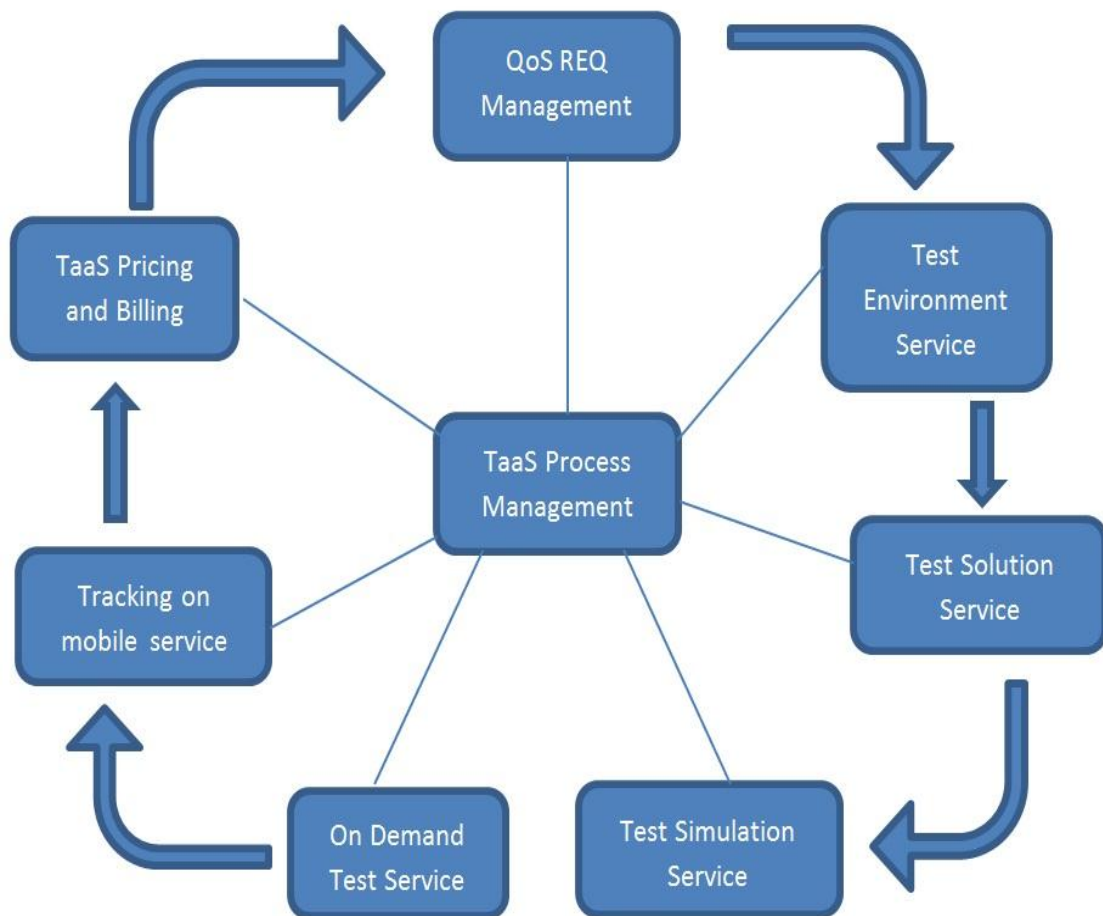


Figure 4: Workflow of TaaS

3.4 Cloud Testing VS. Conventional Software Testing

Table 1 below show comparison between Cloud Testing and conventional software testing based on different parameters.

Parameter	Internet-Based Software Testing (i.e. Distributed/Web-Based System Infrastructure)	Cloud-Based Software Testing
Primary Testing Objectives	Assure the quality of system functions and performance based on the given specifications Check usability, compatibility, interoperability.	Assure the quality of functions and performance of SaaS, Clouds, and applications by leveraging a cloud environment. Assure the quality of cloud elasticity & scalability based a SLA.
Testing as a service	In-house internal software testing as engineering tasks.	Real-time on-demand testing service offered by a third-party. Online testing service based on a pre-defined SLA.
Testing Environment	A pre-fixed and configured test environment in a test lab.	An open public test environment with diverse computing resources. A scalable private test environment in a test lab.
Testing Costs	Required hardware costs and software (license) costs Engineering costs in a test process.	-Based on pre-defined SLA's -Pay as you test (Cloud testing cost) -Engineering cost in SaaS/Cloud/application vendors.
Test simulation	-Simulated online user access -Simulated online traffic data	-Virtual /Online user access simulation -Virtual /Online traffic data simulation
Functional Testing	-Validating functions (unit and system) as well as its features	-SaaS/Cloud service functions -End-to-end application functions
Integration Testing	-Function based -Component based -Architecture based	-SaaS based integration in cloud -SaaS integration between clouds -End to end integration over clouds
Security testing	-Function based security features - User privacy - Client / server based security - Process based security.	-SaaS/Cloud security features, - User privacy in diverse web clients - SaaS/Cloud API and connectivity security - Security testing with virtual /real-time tests in vendor's cloud
Scalability & performance testing	-Performed a fixed test environment - Simulated user access and test data. - Online monitor and evaluation.	-Performed in a scalable test environment based on SLA - Apply both virtual and real time online test data - Online monitor, validate and measurement.

Table 1: Comparison between Cloud testing and Conventional software testing

Cloud based testing model helps overcome many challenges in testing flat form like Cloud offers range of opportunity to take up certain kind of projects, opportunity to reduce cost, optimize delivery times, control processes and systems. Some of the major challenges faced in testing are typically infrastructure setup, establishment of proper tools and processes, availability of skilled test engineers. Cost along with these challenges causes the decision makers to compromise on critical aspects and release software without completely validating their products. Cloud based testing utilizes the cloud computing concept to extend the current testing paradigms using shared, scalable, 'on-demand' testing infrastructure. Cloud based testing model's flexibility reduces capital cost, risk and effort associated with establishing an appropriate testing environment for the enterprise. The next two figures (Figure

5, 6) give you an idea of how different our cloud testing services are and perusing it makes it imperative of claims of saving at least 40% on your testing activities.

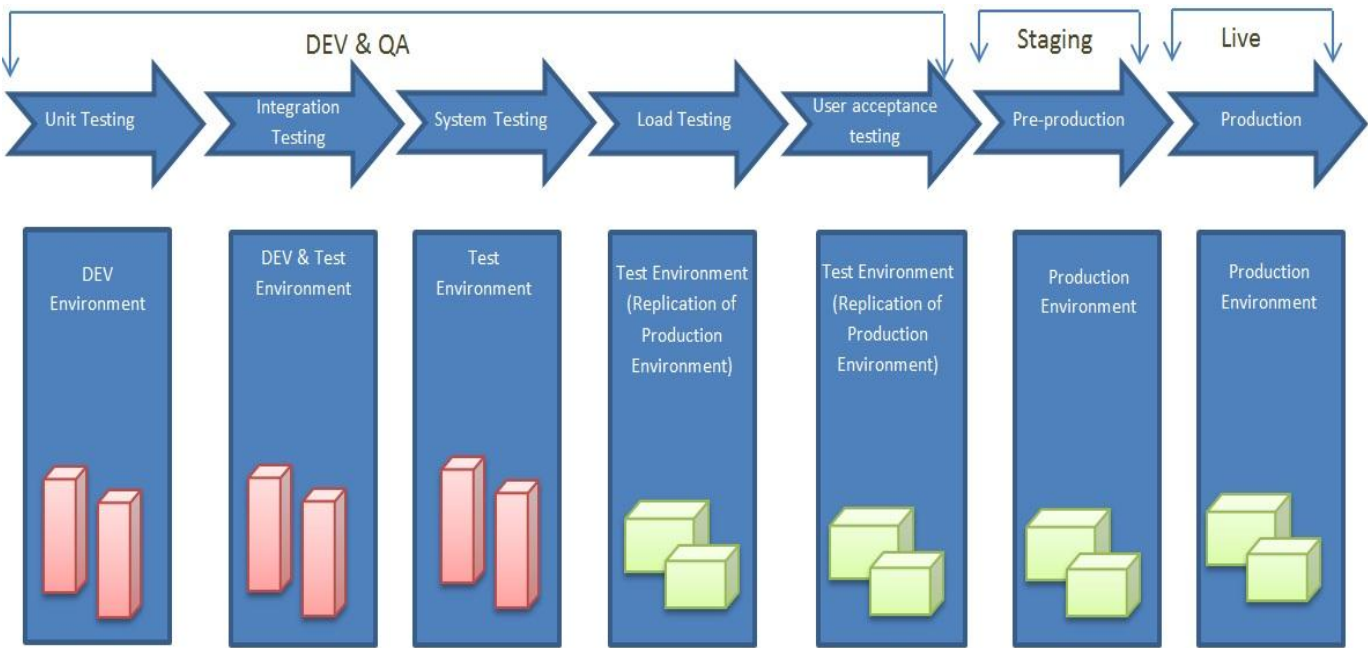


Figure 5: Traditional Software Testing Environment

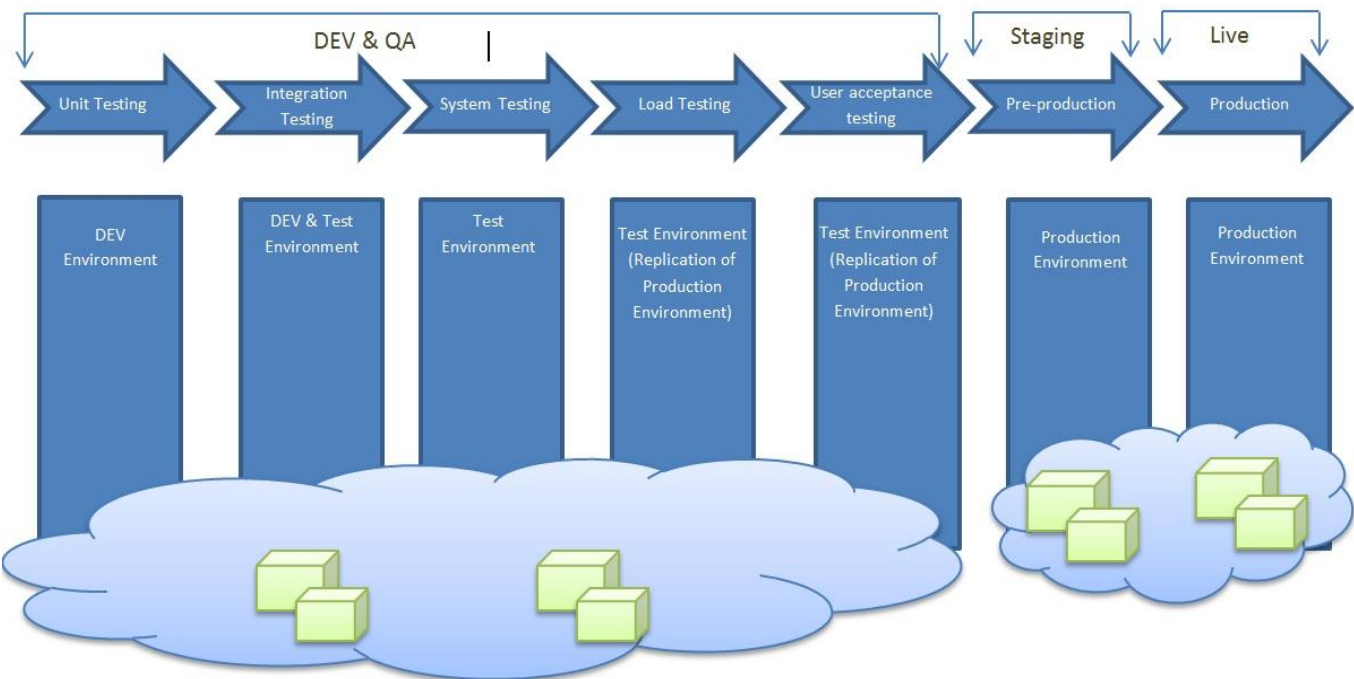


Figure 6: Cloud Based Test Environment

4. TOOLS FOR CLOUD TESTING

Cloud simulators are required for cloud system testing to decrease the complexity and separate quality concerns. They enable performance analysts to analyze system behavior by focusing on quality issues of specific component under different scenarios. Some of the published clouds computing simulators for evaluating cloud computing systems performance are listed below in this section.

- Soatest
- HP LoadRunner
- CloudSim
- CloudAnalyst
- GreenCloud
- NetworkCloudSim
- EMUSIM
- SPECI
- GroudSim

5. MAJOR ISSUE IN CLOUD TESTING

There are a number of major issues; some of them are discussed below:

- ❖ **On-demand test environment construction** – How to set up a testing environment systematically (or automatically) for on-demand testing services in a cloud? Although the current cloud technologies support automatic provision of required computing resources for each SaaS (or application) in a cloud, there are no supporting solutions to assist engineers to set up a required test environment in a cloud using a cost-effective way.
- ❖ **Scalability and performance testing** - Although many published papers discuss system performance testing and scalability evaluation in the past two decades, most of them address issues and solutions in conventional distributed software or web-based software systems. Since these systems are set up with pre-configured system resources and infrastructures, performance testing and scalability evaluation are usually conducted in a static and pre-fixed system environment (such as a test lab.), so the existing evaluation metrics, frameworks, and solutions did not consider the special features in cloud testing, such as dynamic scalability, scalable testing environments, SLA-based requirements, and cost-models.
- ❖ **Testing security and measurement in clouds** – Security testing has becoming a hot research subject with many open questions in current software testing community. Since security becomes a major concern inside clouds and security services become a necessary part in modern SaaS and cloud technology, engineers must deal the issues and challenges in security validation and quality assurance for SaaS and clouds.
- ❖ **Integration testing in clouds** - One of the major reasons is the existing software and components are developed without Enabling technology and solution to support and facilitate systematic software integration. In a cloud infrastructure, engineers must deal with integration of different SaaS and applications in/over clouds in a black-box view based on their provided APIs and connectivity protocols.
- ❖ **On-demand testing issues and challenges** - In TaaS, software testing services must be controlled and managed based on on-demand testing requests. This kind of new testing service model raised several issues and challenges.
- ❖ **Regression testing issues and challenges** - Supporting on-demand software validation in clouds must address the regression testing issues and challenges caused by software changes and bug-fixing. However, most existing research in software regression testing pays most attention to re-test a specific software version in a pre-configured test environment.

6. BENEFITS

The ability and cost to simulate web traffic for software testing purposes has been an inhibitor to overall web reliability. The low cost and accessibility of the cloud's extremely large computing resources provides the ability to replicate real world usage of these systems by geographically distributed users, executing wide varieties of user scenarios, at scales previously unattainable in traditional testing environments. Minimal start-up time along with quality assurance can be achieved by cloud testing.

Following are some of the key benefits:

- ✓ Reduction in capital expenditure
- ✓ Highly scalable

7. CURRENT SCENARIO

An evolving cloud testing community is forming under the auspices of the Software Testing & Quality Assurance group hosted by LinkedIn. Testing professionals openly share their experiences and exchange ideas related to cloud testing in order to enhance each other's proficiencies.

8. CONCLUSION AND FUTURE WORK

Cloud testing is becoming a hot research topic in cloud computing and software engineering community. As the advance of cloud technology and testing as services, more research work must be done to address the open issues and challenges in cloud testing and TaaS. More innovative testing techniques and solutions, and quality of service (QoS) standards are needed to support on-demand testing services in a scalable cloud infrastructure. The paper includes discussion about cloud testing in terms of its special requirements, benefits, and features as well as the comparison with conventional testing.

9. REFERENCES

- [1] J. Hurwitz, M. Kaufman, and R. Bloor, "Cloud Computing for Dummies," Wiley Publishing, Inc. 2010.
- [2] T. Vengattaraman, P. Dhavachelvan, R. Baskaran, "Model of Cloud Based Application Environment for Software Testing," (IJCSIS) International Journal of Computer Science and Information Security, Vol. 7, No. 3, 2010.
- [3] Leah Muthoni Riungu, Ossi Taipale, Kari Smolander, "Software Testing as an Online Service: Observations from Practice," In Third International Conference on Software Testing, Verification, and Validation Workshops (ICSTW), 418-423, 2010.
- [4] Tauhida Parveen, Scott Tilley, "When to Migrate Software Testing to the Cloud?," In Third International Conference on Software Testing, Verification, and Validation Workshops (ICSTW), 424-427, 2010.
- [5] W. K. Chan, Lijun Mei, Zhenyu Zhang, "Modeling and Testing of Cloud Applications," In 2009 IEEE Asia-Pacific Service Computing Conference (APSCC 2009), Singapore; Dec 7-11, 2009.
- [6] Liu Gu, Shing-Chi Chenug, "Constructing and testing privacy-aware services in a cloud computing environment: challenges and opportunities," In the First Asia-Pacific Symposium on Internetware, Beijing, China, 2009.
- [7] George Candea, Stefan Bucur, Zamfir Cristian, "Automated Software Testing as a Service (TaaS)," In the 1st ACM Symposium on Cloud Computing, 2010.
- [8] H. Lu, W. K. Chan, and T. H. Tse., "Testing pervasive software in the presence of context inconsistency resolution services," In the 30th International Conference on Software Engineering (ICSE 2008), 61-70, 2008.
- [9] G. Goth, "Googling test practices? Web giant's culture encourages process improvement," IEEE Software , vol. 25, no. 2, 92-94, 2008.
- [10] R. Collard, "Performance innovations, testing implications", Software Test & Performance Magazine, Vol. 6, No. 8, 19-20, August 2009.
- [11] "IBM Smart Business Development and Test Cloud," IBM Global Technology Services, 2010. URL: <http://www935.ibm.com/services/us/index.wss/offering/middleware/a1030965>.
- [12] B. Wrenn, CISSP, ISSEP, "Unisys Secure Cloud Addressing the Top Threats of Cloud Computing," (white paper).

- [13] P. Jogalekar, M. Woodside. "Evaluating the scalability of distributed systems," IEEE Trans. Parallel and Distributed Systems, vol. 11, no. 6, 589–603, 2000.
- [14] Wikipedia, the free encyclopedia "Cloud testing", http://en.wikipedia.org/wiki/Cloud_testing
- [15] www.ijera.com, "ISSN: 2248-9622", "International Journal of Engineering Research and Applications", Vol. 2, Issue 4, July-August 2012,
- [16] TYSS's cloud based testing "<http://www.testyantra.com/cloud-testing>"

Cloud Testing